Response to the Editor and Reviewers

Manuscript: LLM-guided headline rewriting for clickability enhancement without clickbait

We thank the editor and both reviewers for their careful and constructive reports. We have expanded the experimental program, converting a manuscript whose central claims rested on qualitative examples into one in which every claim is backed by quantitative evidence with independent metrics, item-matched significance testing, and confidence intervals. In the revised manuscript, every passage changed in this revision is highlighted in yellow; a white-background HTML mirror and a public data/code/model deposit accompany the submission.

The revised Section 4 (Experiments and Results) now reports, all on held-out data and with independent evaluators wherever possible:

- **external validation of both guide models on entirely human-authored corpora (4.5);**
- *prefix-level validation of the guide on real headlines (4.6, Figure 5);*
- **a quantified Reuters-neutrality check of the source pool (4.7);**
- *a four-cell dual-guidance ablation on 300 held-out headlines (4.8, Table 6, Figure 6);*
- *a comparison against the DExperts and GeDi decoding-time baselines showing that dual control decouples engagement from clickbait (4.9, Table 7, Figure 7, Table 8);*
- *a rater-based evaluation with three independent annotators (4.10, Table 9);*
- *a robustness and factuality analysis (4.11, Table 10);*
- *cross-domain generalization to non-Reuters headlines (4.12, Table 11);*
- *transfer to a second base-model family, Qwen2.5-7B (4.13, Table 12).*

Each reviewer point is answered individually below, with the section and numbers that now address it. Part 2 lists improvements we made that were not explicitly requested.

Part 1 — Point-by-point responses

Reviewer 1

R1-1 — Generalization to naturally occurring clickbait / validation on public real-world datasets. Addressed (Sections 4.5–4.6). Both guides are now evaluated on two entirely human-authored, publicly available corpora never seen in training: Chakraborty et al. 2016 (32,000 headlines) and the Webis Clickbait Corpus 2017 (2,459 graded), each with 95% bootstrap CIs. The clickbait guide reaches AUROC 0.955 on Chakraborty and 0.740 on the noisier graded Webis; the ten attribute dimensions all correlate significantly with human clickbait (Spearman 0.34–0.45, every $p < 1\times10^{-60}$). A new prefix-level test (Section 4.6, Figure 5) shows the guide predicts the human label from an early prefix, with AUROC rising monotonically from 0.891 at 30% of the headline to 0.954 at full length. Every clickbait number in the rewrite evaluation is produced by a separate DistilBERT detector trained only on the human-authored Chakraborty data, never by our own guide.

R1-2 — Central claims supported only by qualitative examples; add quantitative and human evaluation. Addressed (Sections 4.8, 4.10). Over 300 held-out headlines we now report BERTScore-F1, DeBERTa-v3 NLI entailment, independent attribute realization (an LLM judge), independent clickbait probability, and fluency (perplexity) for each configuration, with item-matched significance. We additionally conducted a rater-based evaluation (Section 4.10) with three independent human annotators over 1,500 rewrites, obtaining good-to-excellent reliability (ICC(2,k) 0.82–0.94); annotator faithfulness correlates with automatic BERTScore (Spearman $\rho = 0.61$, $n = 1500$). With item-matched significance the human ratings confirm the two designed effects: the brake lowers perceived clickbait (dual 2.03 vs no-guidance 2.61 on a 1–5 scale, $p = 1\times10^{-7}$) and the dual guidance is rated more faithful than the no-guidance baseline (3.58 vs 2.89, $p = 1\times10^{-7}$) and than the DExperts and GeDi baselines (both $p < 1\times10^{-13}$). On engagement we are explicit about a tradeoff: the human 1–5 rating is lower at the dual operating point than at no-guidance (2.90 vs 3.46), because perceived click-appeal is entangled with clickbait cues. Engagement is nonetheless controllable, not lost: the positive guide raises perceived engagement on its own (engage-only 3.51 vs 3.46), and an independent LLM engagement judge scored across the α–β grid ($n = 840$) holds engagement at or above the no-guidance baseline at the operating point (0.925 vs 0.900) while cutting clickbait, with a lower-brake point (4, 1) higher still — so engagement is a controllable axis set by α (Section 4.10). The full annotation workbooks and per-item ratings are released in the deposit.

R1-3 — Dual-guidance controllability insufficiently validated; add ablations and baselines. Addressed (Sections 4.8–4.9). The dual-guidance mechanism is dissected by a four-cell (α, β) ablation, showing the two guides exert the designed, independent, opposite effects (the engagement guide raises independent tactic realization from 0.48 to 0.51; the brake lowers the independent clickbait score from 0.079 to 0.057). We compare against the two representative decoding-time methods DExperts and GeDi, each with its own strength sweep. The dual guide adds an independent clickbait brake (Table 7, Figure 7) that the single-axis methods lack: a lever that lowers induced clickbait separately from the engagement weight. Because their single control couples engagement and clickbait, the baselines reach the tactic at low clickbait only at low steering; any stronger steering drives their induced clickbait up steeply (DExperts 0.06 to 0.73 across its sweep; GeDi to 0.59) with fidelity falling in step, whereas the brake holds induced clickbait low across the range. For a fair comparison DExperts is decoded with a repetition penalty (1.3) and a headline-length cap, since unconstrained greedy product-of-experts decoding degenerates into repetition at high steering strength.

Reviewer 2

R2-M1 — Unsupported intro claims; ungrounded, unvalidated rubric. Addressed. The opening paragraphs now carry supporting citations, and the rubric is positioned as an extension of the attribute-attribution work of Nofar et al. 2025. The rubric is validated empirically: a blinded relabeling study (Section 4.10, C1) over 150 headlines yields substantial inter-rater agreement (mean Fleiss $\kappa = 0.70$).

R2-M2 — Central claims not supported by quantitative evidence. Addressed. Every claim in the Conclusion is now tied to a measured result in Section 4: the brake measurably lowers both the independent detector score and human-rated clickbait (dual 2.03 vs 2.61, $p = 1\times10^{-7}$), independent human annotators rate the dual guidance more faithful than the no-guidance baseline (3.58 vs 2.89, $p = 1\times10^{-7}$), and against the baselines the dual control holds induced clickbait low at matched tactic realization while the single-axis methods' clickbait rises steeply with steering.

R2-M3 — Circularity of the evaluation loop. Addressed (Section 4.5). Both guides are re-evaluated on entirely human-authored corpora, scoring AUROC 0.955 / 0.740 there alongside the in-distribution synthetic score (0.9998). Every clickbait number in the rewrite evaluation uses an independent DistilBERT detector trained only on human data, which breaks the loop the reviewer identified.

R2-M4 — No baselines or ablations. Addressed (Sections 4.8–4.9, Tables 6, 7, 8; Figures 6, 7). Both are now present: the full four-cell ablation and the DExperts and GeDi baselines, with a decoupling analysis (Table 7, Figure 7) and a qualitative side-by-side (Table 8) showing that when the single-axis baselines are pushed to high steering their induced clickbait rises steeply, whereas the dual guide's independent brake keeps it low. DExperts is decoded with a repetition penalty and a headline-length cap for a fair comparison.

R2-M5 — Missing implementation details / reproducibility. Addressed (Section 3.6). The base model (Meta-Llama-3-8B-Instruct), top-k = 50, the log-domain objective and its swept weights, the tuning grid, cloud hardware, guide-training settings, the full synthetic-data generation procedure (batch size, tactic sampling, JSON self-repair), and per-headline wall-clock are all specified. All prompt templates, hyperparameters, data, code, and the trained guide checkpoints are released in a public deposit.

R2-M6 — Apparent Table 2 inconsistencies; validate tactic-label fidelity. Addressed (Section 4.10, C1). A blinded rubric re-labeling yields $\kappa = 0.70$ and a mean intended-vs-consensus tactic F1 of 0.44, matching the attribute model's own recoverability and reflecting the graded, fine-grained nature of the tactics. The synthetic examples in Table 2 were also refreshed from the authoritative generation run.

R2-M7 — Attribute model may be too weak to guide reliably. Addressed (Sections 4.8, 3.7). The $(0,0)\rightarrow(4,0)$ contrast isolates the positive guide's contribution directly, and attribute realization is measured by an independent LLM judge rather than the guide itself, so classifier-noise propagation is quantified rather than assumed. The judge's own reliability is cross-checked against the rater study's recoverability (Section 3.7).

R2-M8 — Some controlled outputs contradict the framework's goals. Addressed (Section 4.11, Table 10). Each boundary case is discussed explicitly, and the embellishment concern is quantified: a stricter new-fact judge (validated on faithful-paraphrase controls that score ≈ 0.02 and fact-injected controls that score 1.0, AUROC ≈ 1.0) shows the dual operating point introduces no more genuine new facts than an unguided rewrite (0.20 vs 0.18 for the no-guidance baseline, not significant), so the guidance does not trade faithfulness for engagement. The apparent embellishment is the rhetorical-question form, not fabricated content.

R2 (review question) — "No statistical analysis (single runs, no variance, no significance testing)." Addressed throughout. Every reported comparison uses item-matched Wilcoxon signed-rank tests; all table estimates carry 95% percentile-bootstrap confidence intervals, now including the single-axis tactic values in Table 7. Matched-pairs rank-biserial effect sizes are reported for the key contrasts — the ablation Paired-win column of Table 6 (r = 0.19–0.22) and the rater faithfulness contrasts (dual vs no-guidance / DExperts / GeDi, r = 0.67 / 0.82 / 0.91), collected in Section 3.7. Holm correction is applied within two explicit families (the four ablation contrasts against the (0,0) cell; the cross-method rater faithfulness comparisons), and every reported significant p-value remains significant after correction within its family (Section 3.7).

R2-m1 — Cross-reference errors. Fixed. The methodology subsections referenced in the introduction now exist and resolve correctly.

R2-m2 — Reuters-neutrality assumption unverified. Addressed (Section 4.7). The independent detector was run over the full 20,825-headline source pool: only 0.61% exceed the clickbait threshold, confirming the neutral baseline rather than assuming it. The domain scope is stated explicitly in the manuscript.

R2-m3 — Is the dataset split stratified? Clarified (Sections 4.2–4.3). The split is performed uniformly at random at the source-headline level (not stratified by tactic); because every source contributes both a neutral and a clickbait sample, class balance is preserved by construction.

R2-m4 — Figure y-axis / caption mismatch. Fixed. Figure 3 now labels the axis and caption consistently; the per-tactic figure was regenerated from the current guide.

Part 2 — Additional improvements (not explicitly requested)

Beyond the reviewers' requests, we strengthened the manuscript as follows; all are highlighted in the manuscript.

1. *Full statistical protocol. Bootstrap CIs, effect sizes, and multiple-comparison correction are applied uniformly across Section 4, not only where variance was questioned.*

2. *Cross-domain generalization (Section 4.12, Table 11). The same guides, at the same operating point, reproduce their designed effects on 150 human-authored non-Reuters headlines (a different register): the brake lowers independent clickbait (0.097 → 0.061) and fidelity is preserved (BERTScore 0.22–0.28), extending the demonstrated scope to a second register. Because those source headlines are drawn from the Chakraborty corpus on which the DistilBERT evaluator was trained, we further retrained the detector from scratch with those 150 sources excluded (held-out accuracy 0.99) and re-scored the rewrites: the brake still lowers induced clickbait (0.031 → 0.025 with engagement, 0.017 → 0.011 without) and engagement still raises it (0.017 → 0.031), so the directional effect is not an artifact of source leakage into the evaluator.*

3. *Transfer to a second base generator (Section 4.13, Table 12). We frame this as a model-family transfer check rather than a second demonstration of the decoupling: the same guides and operating point run on Qwen2.5-7B-Instruct with source fidelity and fluency preserved (BERTScore and perplexity at or better than the Llama-3 configuration). The Table 12 tactic column now reports the independent LLM judge (replacing the guide-based circular score); on Qwen it is flat across cells (≈ 0.50), so the sharp brake-induced drop seen under the circular score was an artifact, not a real loss. Because Qwen's rewrites sit near the detector floor throughout, the decoupling itself is established on the Llama results (Table 7, Figure 7), not re-demonstrated here. For multilingual base generators we add a language-consistency constraint to the decoder (Section 3.6).*

4. *Factuality calibration (Table 10). The new-fact judge and NLI entailment are calibrated on labelled controls, giving the fidelity analysis a verified basis.*

5. *Qualitative depth. A worked "engagement continuum" for one headline across the α sweep (Section 4.4), a full four-cell grid on three headlines (Table 4), and a qualitative cross-method comparison (Table 8) make the control behaviour concrete.*

6. *Objective and evaluators. The combined objective uses the canonical log-domain FUDGE factorization; all evaluators (independent DistilBERT detector, LLM tactic judge, DeBERTa-v3 NLI, distilGPT-2 perplexity) are independent of the guides.*

7. *Complete citation coverage. All datasets, methods, models, and evaluation metrics are now cited, including the base generators (Llama-3, Qwen2.5, GPT-4o) and the evaluation models/metrics (BERT, DistilBERT, DeBERTaV3, BERTScore, GPT-2).*

8. *Reproducibility deposit. Data, code, and the trained guide checkpoints are released publicly, with per-item score files behind every table.*

9. *Presentation. Consistent Scientific Reports reference style, a white-background HTML mirror, and full yellow highlighting of all revisions.*

Each central claim is now backed by the data and independent evaluators described above. We thank the reviewers again for feedback that strengthened the work.

Sincerely, The authors